

Computing Mixture Models with EM

- Given n objects $O = \{o_1, \dots, o_n\}$, we want to find a set of parameters $\Theta = \{\theta_1, \dots, \theta_k\}$ such that $P(O|\Theta)$ is maximized, where $\theta_j = (\mu_j, \sigma_j)$ are the mean and standard deviation of the j -th univariate Gaussian distribution
- We initially assign random values to parameters θ_j , then iteratively conduct the E- and M- steps until converge or sufficiently small change

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- At the E-step, for each object o_i , calculate the probability that o_i belongs to each distribution, $P(\Theta_j | o_i, \Theta) = \frac{P(o_i | \Theta_j)}{\sum_{l=1}^k P(o_i | \Theta_l)}$
- At the M-step, adjust the parameters $\theta_j = (\mu_j, \sigma_j)$ so that the expected likelihood $P(\mathbf{O} | \Theta)$ is maximized

$$\mu_j = \sum_{i=1}^n \frac{P(\Theta_j | o_i, \Theta)}{\sum_{l=1}^n P(\Theta_j | o_l, \Theta)} = \frac{1}{k} \frac{\sum_{i=1}^n o_i P(\Theta_j | o_i, \Theta)}{\sum_{l=1}^n P(\Theta_j | o_l, \Theta)}$$

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^n P(\Theta_j | o_i, \Theta) (o_i - \mu_j)^2}{\sum_{i=1}^n P(\Theta_j | o_i, \Theta)}}$$

Advantages and Disadvantages of Mixture Models

□ Strength

- Mixture models are more general than partitioning and fuzzy clustering
- Clusters can be characterized by a small number of parameters
- The results may satisfy the statistical assumptions of the generative models

□ Weakness

- Converge to local optimal (overcome: run multi-times with random initialization)
- Computationally expensive if the number of distributions is large, or the data set contains very few observed data points
- Need large data sets
- Hard to estimate the number of clusters